Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- jqassistant
- Neo4j Python Driver

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	58987	52.261471
1	[Git, Change, Create]	17329	15.353197
2	[Git, Commit]	11316	10.025782
3	[Git, Change, Delete]	8787	7.785131
4	[File, Git]	5921	5.245905
5	[Git, Change, Rename]	3354	2.971587
6	[Git, Tag]	1795	1.590339
7	[Author, Git, Person]	1268	1.123426
8	[Json, Key]	668	0.591837
9	[Json, Value, Scalar]	603	0.534248
10	[Committer, Git, Person]	370	0.327814
11	[NPM, Dependency]	338	0.299462
12	[Type, TS, Primitive]	293	0.259593
13	[Type, TS, Declared]	277	0.245417
14	[TS, ExternalDeclaration]	215	0.190486
15	[Git, Change, Copy]	138	0.122266
16	[Type, TS, Literal]	136	0.120494
17	[Json, Value, Object]	133	0.117836
18	[Type, TS, Union]	120	0.106318
19	[Type, TS, ObjectMember]	103	0.091256
20	[NPM, Script]	91	0.080624
21	[TS, Property]	65	0.057589
22	[TS, Function]	47	0.041641
23	[Git, Branch]	46	0.040755
24	[Type, TS, FunctionParameter]	40	0.035439
25	[Type, Object, TS]	39	0.034553
26	[File, Directory]	34	0.030123
27	[Type, TS, Function]	34	0.030123
28	[TS, Parameter]	33	0.029237
29	[Package, File, Json, NPM]	29	0.025694

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

```
<Figure size 640x480 with 0 Axes>
```

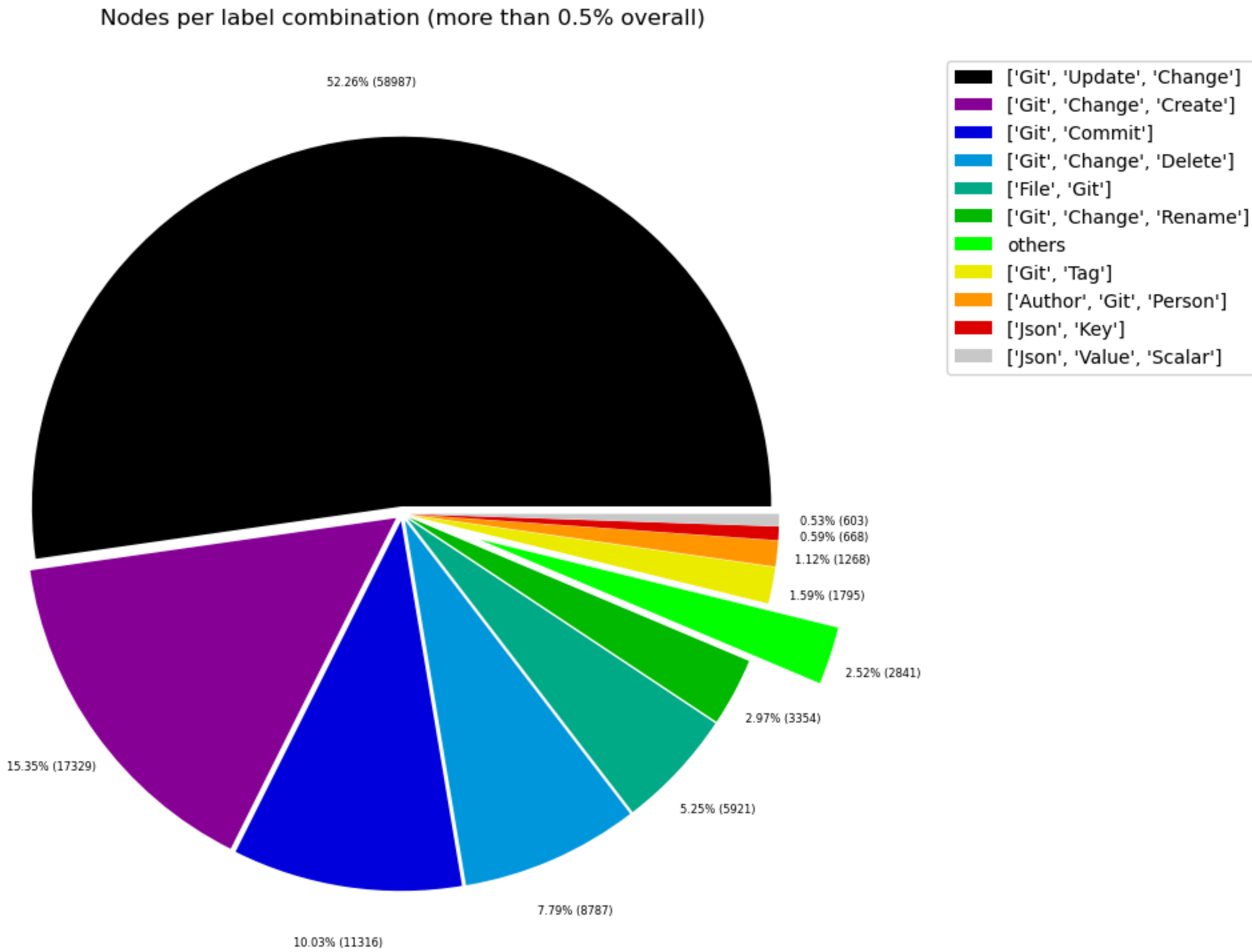

Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000886
1	[NPM, PublishConfig]	1	0.000886
2	[File, TS, Scan]	1	0.000886
3	[TS, Method]	1	0.000886
4	[Repository, File, Git]	1	0.000886
5	[TS, Constructor]	1	0.000886
6	[Value, TS, ObjectMember]	1	0.000886
7	[TS, Class]	1	0.000886
8	[TS, Enum]	2	0.001772
9	[Value, Object, TS]	3	0.002658
10	[Type, TS, Tuple]	3	0.002658
11	[Value, TS, Function]	4	0.003544
12	[TS, TypeParameter]	4	0.003544
13	[Value, TS, Complex]	5	0.004430
14	[Repository, NPM]	5	0.004430
15	[NPM, Engine]	6	0.005316
16	[Project, TS]	6	0.005316
17	[File, Local]	6	0.005316
18	[Type, TS, TypeParameterReference]	6	0.005316
19	[Value, TS, Member]	6	0.005316
20	[File, TS, Local, Module]	6	0.005316
21	[Value, TS, Call]	6	0.005316
22	[TS, EnumMember]	8	0.007088
23	[Type, TS, NotIdentified]	11	0.009746
24	[TS, ExternalModule]	11	0.009746
25	[Json, Value, Array]	12	0.010632
26	[Value, TS, Declared]	13	0.011518
27	[TS, TypeAlias]	16	0.014176
28	[File, Directory, Local]	16	0.014176
29	[TS, Interface]	17	0.015062

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```

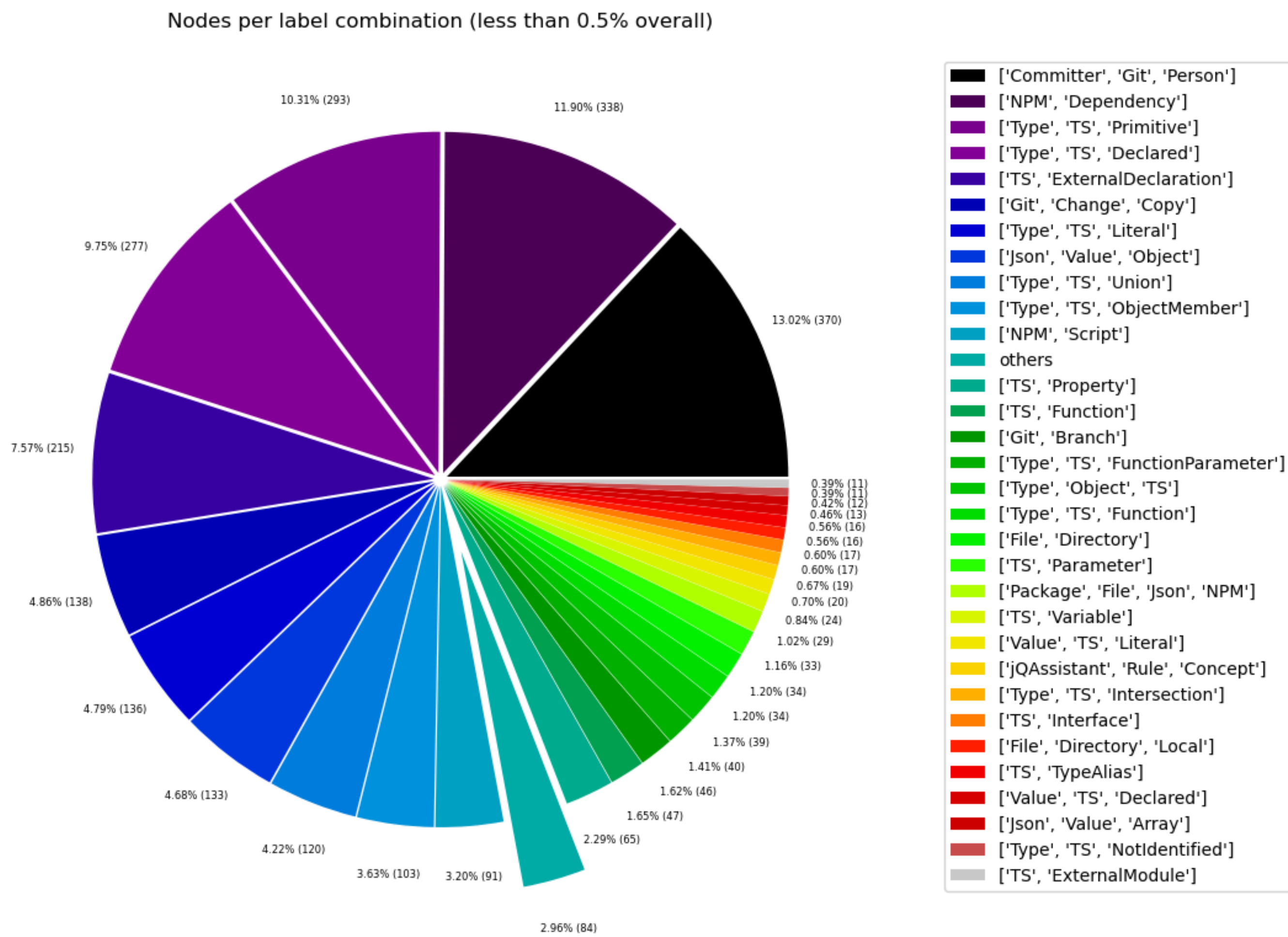

Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	109312	96.848559
1	Change	88595	78.493652
2	Update	58987	52.261471
3	Create	17329	15.353197
4	Commit	11316	10.025782
5	Delete	8787	7.785131
6	File	6014	5.328301
7	Rename	3354	2.971587
8	Tag	1795	1.590339
9	Person	1638	1.451240
10	TS	1595	1.413143
11	Json	1445	1.280245
12	Author	1268	1.123426
13	Type	1079	0.955976
14	Value	806	0.714102
15	Key	668	0.591837
16	Scalar	603	0.534248
17	NPM	470	0.416412
18	Committer	370	0.327814
19	Dependency	338	0.299462
20	Primitive	293	0.259593
21	Declared	290	0.256935
22	ExternalDeclaration	215	0.190486
23	Object	175	0.155047
24	Literal	156	0.138213
25	Copy	138	0.122266
26	Union	120	0.106318
27	ObjectMember	104	0.092142
28	Script	91	0.080624
29	Function	85	0.075309
30	Property	65	0.057589
31	Directory	50	0.044299
32	Branch	46	0.040755
33	FunctionParameter	40	0.035439
34	Parameter	33	0.029237
35	Package	29	0.025694
36	Local	28	0.024808
37	Variable	24	0.021264
38	jQAssistant	20	0.017720
39	Concept	19	0.016834

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

```
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```
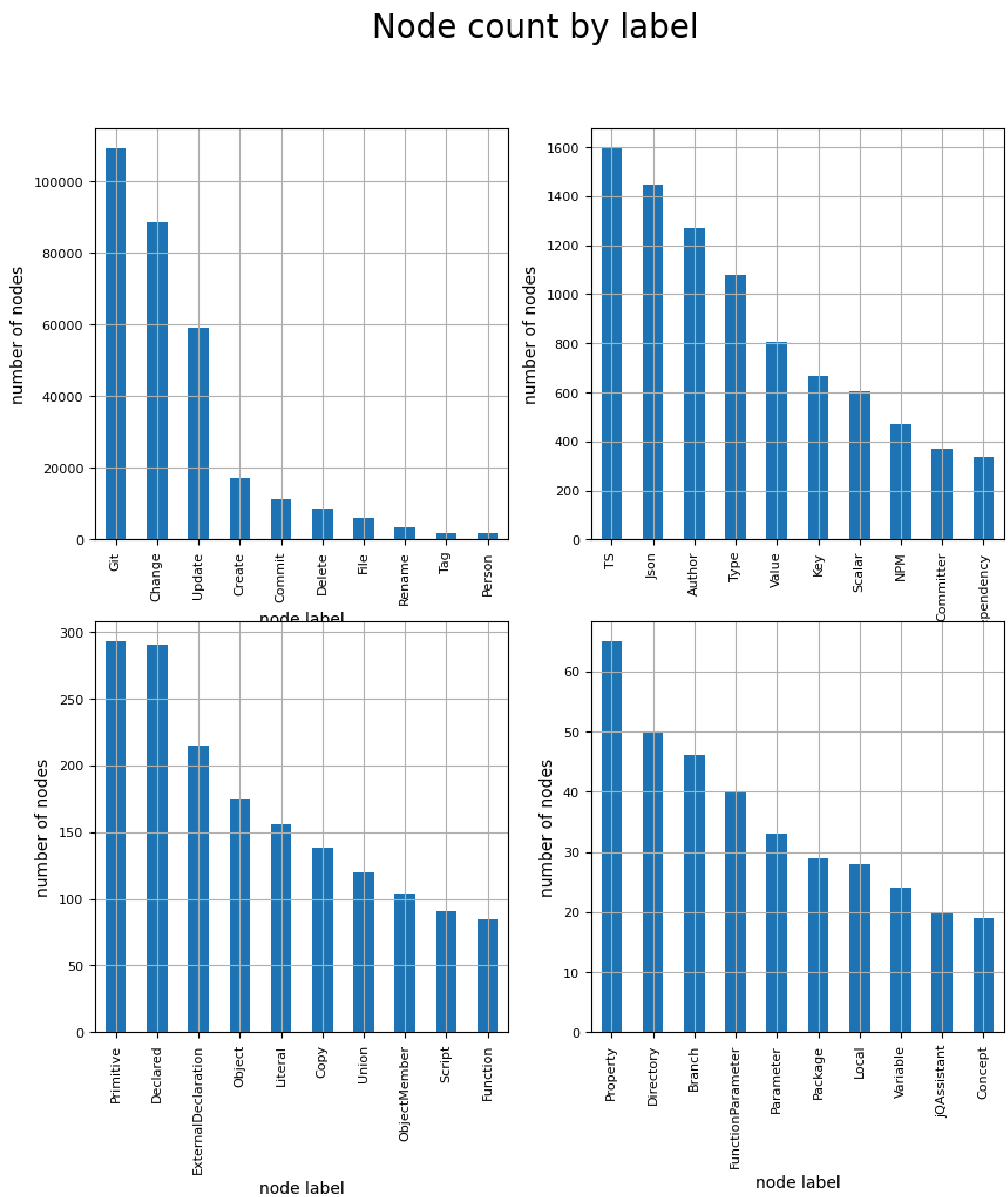



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

```
Total number of relationships: 336984
```

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	88595	26.290566
1	MODIFIES	88595	26.290566
2	UPDATES	58987	17.504392
3	COMMITTED	22632	6.716046
4	CREATES	20821	6.178632
5	HAS_PARENT	12402	3.680293
6	DELETES	12141	3.602842
7	HAS_COMMIT	11316	3.358023
8	HAS_FILE	5921	1.757057
9	RENAMES	3354	0.995299
10	HAS_TAG	1795	0.532666
11	ON_COMMIT	1795	0.532666
12	HAS_NEW_NAME	1770	0.525247
13	HAS_AUTHOR	1268	0.376279
14	DEPENDS_ON	887	0.263217
15	HAS_KEY	668	0.198229
16	HAS_VALUE	668	0.198229
17	CONTAINS	596	0.176863
18	HAS_COMMITTER	370	0.109797
19	OF_TYPE	339	0.100598
20	EXPORTS	309	0.091696
21	REFERENCES	197	0.058460
22	DECLARES	186	0.055195
23	DECLARES_DEV_DEPENDENCY	169	0.050151
24	DECLARES_DEPENDENCY	161	0.047777
25	COPIES	138	0.040951
26	HAS_MEMBER	104	0.030862
27	HAS_TYPE_ARGUMENT	94	0.027894
28	DECLARES_SCRIPT	91	0.027004
29	RETURNS	82	0.024333

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

```
<Figure size 640x480 with 0 Axes>
```

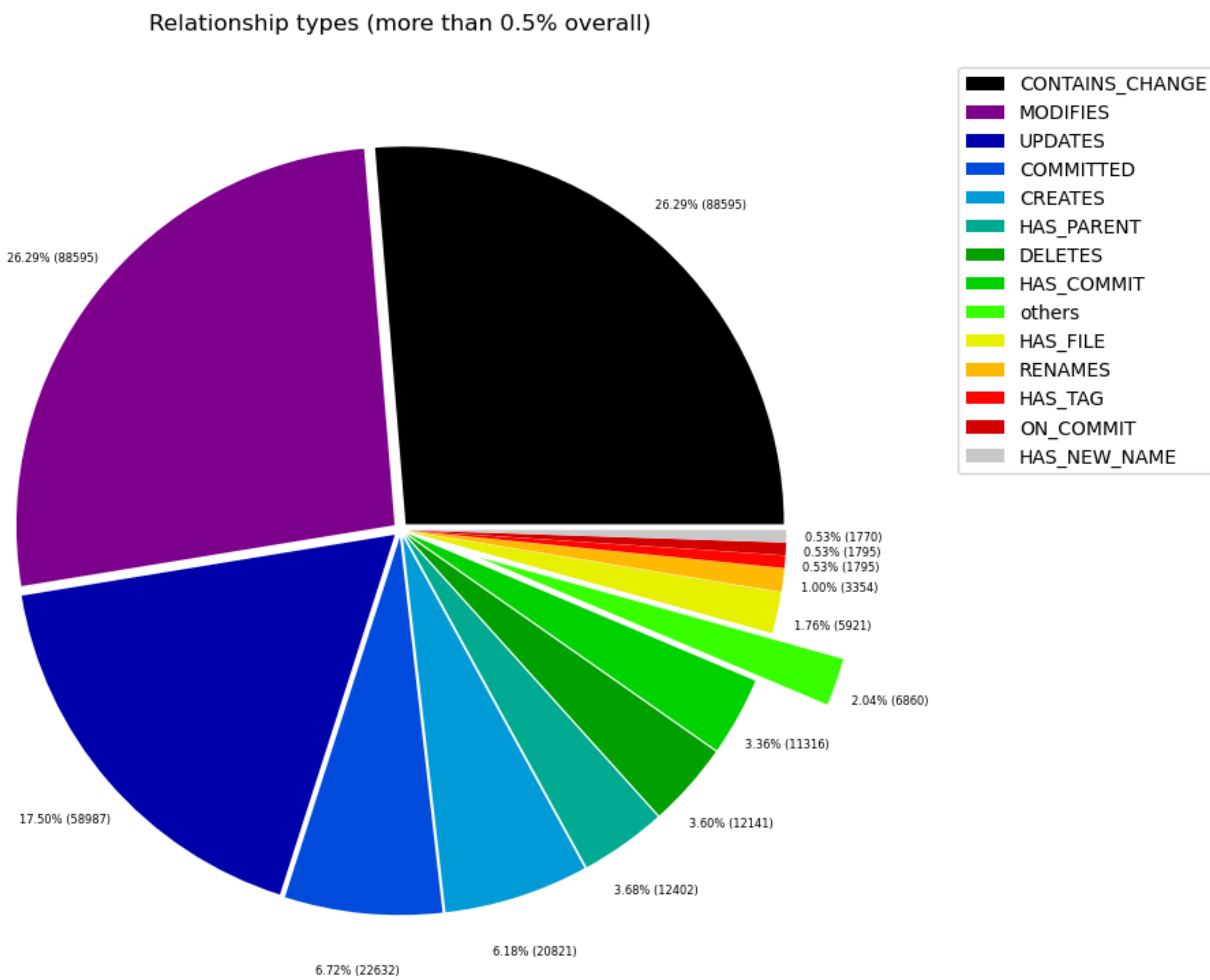


Table 2b - Lowest relationship count by type

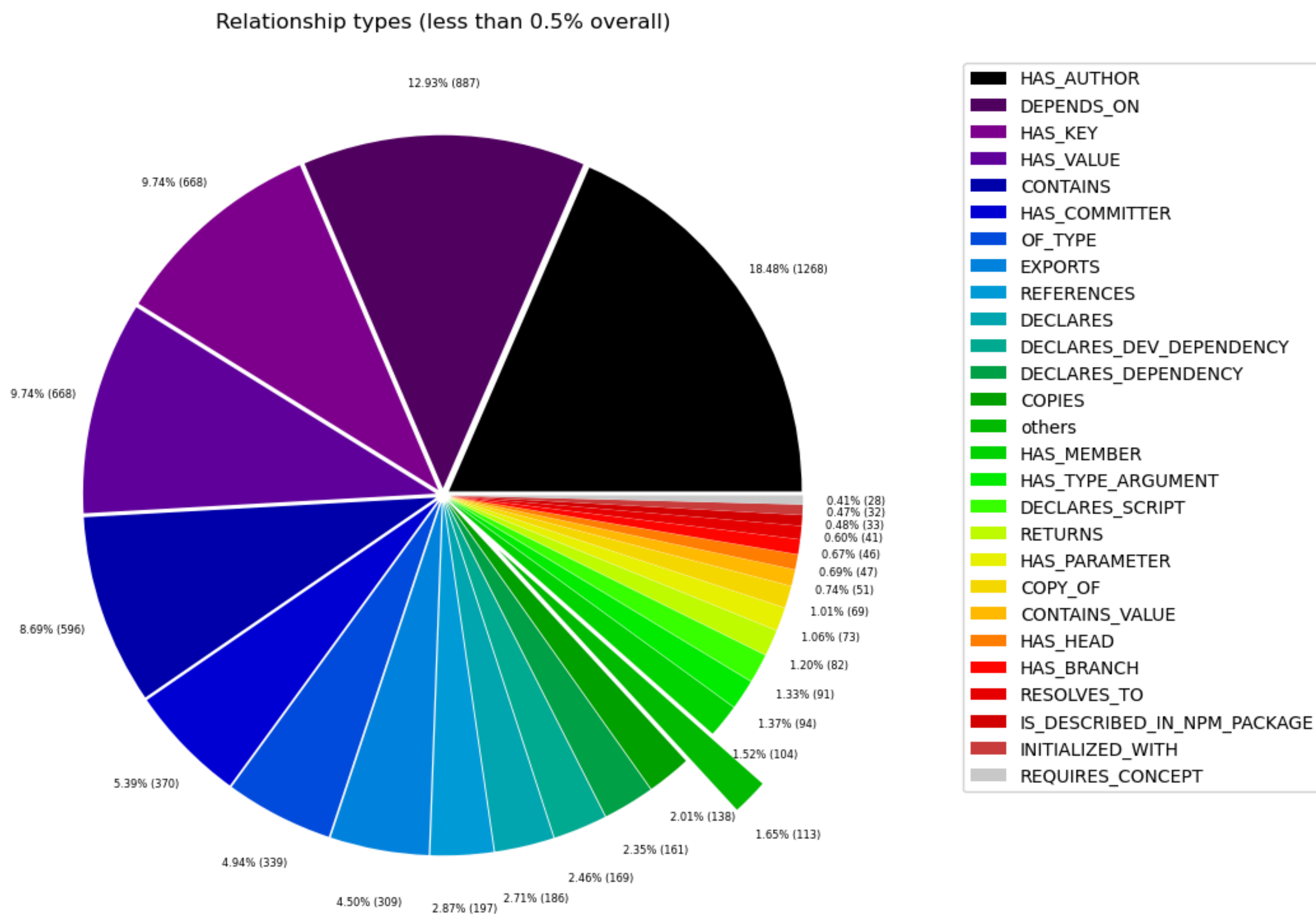
Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000297
1	CONSTRAINED_BY	4	0.001187
2	IN_REPOSITORY	5	0.001484
3	REFERENCED_PROJECTS	5	0.001484
4	PARENT	6	0.001780
5	MEMBER	6	0.001780
6	HAS_ROOT	6	0.001780
7	HAS_NPM_PACKAGE	6	0.001780
8	HAS_CONFIG	6	0.001780
9	HAS_ARGUMENT	6	0.001780
10	EXTENDS	6	0.001780
11	DECLARES_ENGINE	6	0.001780
12	CONTAINS_PROJECT	6	0.001780
13	CALLS	6	0.001780
14	DECLARES_PEER_DEPENDENCY	8	0.002374
15	USES	11	0.003264
16	INCLUDES_CONCEPT	19	0.005638
17	REQUIRES_CONCEPT	28	0.008309
18	INITIALIZED_WITH	32	0.009496
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009793
20	RESOLVES_TO	41	0.012167
21	HAS_BRANCH	46	0.013650
22	HAS_HEAD	47	0.013947
23	CONTAINS_VALUE	51	0.015134
24	COPY_OF	69	0.020476
25	HAS_PARAMETER	73	0.021663
26	RETURNS	82	0.024333
27	DECLARES_SCRIPT	91	0.027004
28	HAS_TYPE_ARGUMENT	94	0.027894
29	HAS_MEMBER	104	0.030862

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	58987	58987	
1	[Git, Update, Change]	UPDATES	[File, Git]	58987	58987	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	58987	11316	
3	[Git, Change, Create]	MODIFIES	[File, Git]	17329	17329	
4	[Git, Change, Create]	CREATES	[File, Git]	17329	17329	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17329	11316	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12402	11316	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11316	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11316	1268	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11316	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8787	11316	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8787	8787	
12	[Git, Change, Delete]	DELETES	[File, Git]	8787	8787	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5921	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3354	11316	
15	[Git, Change, Rename]	DELETES	[File, Git]	3354	3354	
16	[Git, Change, Rename]	RENAMES	[File, Git]	3354	3354	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3354	3354	
18	[Git, Change, Rename]	CREATES	[File, Git]	3354	3354	
19	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1795	1	
20	[Git, Tag]	ON_COMMIT	[Git, Commit]	1795	1795	
21	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5921	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1268	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

```
total_number_of_nodes (vertices): 112869
total_number_of_relationships (edges): 336984
-> total directed graph density: 2.6452320392844444e-05
-> total directed graph density in percent: 0.0026452320392844446
```